Iron Ore Grade Estimation Using Visible–Near-Infrared Hyperspectral Imaging and Convolutional Neural Network

Narihiro Owada, Brian Bino Sinaice, Kazuki Tatsuta, Hiroya Takeuchi, Hisatoshi Toriya, Natsuo Okada, Atsushi Shibayama, and Tsuyoshi Adachi

***Abstract*—Hyperspectral imaging has been widely used in geoscience from space satellites to laboratory equipment due to its huge potential in qualifying and quantifying minerals. However, hyperspectral imaging has also experienced difficulties in detailed ore grade estimations due to its spectral complexity. To address this complexity, this article presents an iron ore grade prediction method that combines hyperspectral imaging and a deep-learning-based convolutional neural network (CNN) prediction model. Hyperspectral images of different grades of iron ore were acquired, followed by training the prediction model using these hyperspectral images. The CNN model was trained using 100 000 spectra. Posttraining prediction results show that the CNN model outperformed conventional models in all evaluation metrics which were 0.974, 2.714, and 11.226 in R^2, validation root mean-squared error (RMSE), and test RMSE, respectively. The analysis results of the trained CNN model furthermore suggested that the model successfully extract spectral features. Finally, the results demonstrate an outstanding prediction capability for the iron ore grade using a CNN and visible–near-infrared (VNIR) hyperspectral imaging for the artificially prepared iron ore samples in the laboratory. Hence, the CNN-based model offers the potential in hyperspectral imaging to efficiently predict ore grade in geological remote sensing applications.**

***Index Terms*—Convolutional neural network (CNN), deep learning, hyperspectral imaging, iron ore, visible–near-infrared (VNIR).**

I. Introduction

HYPERSPECTRAL imaging is a remote sensing technique widely used in geoscience. This imaging technique simultaneously acquires spatial information, that is, the camera's field of view, and spectral information, which is uniquely dependent on materials [1]. Spectra are defined as a function of an electromagnetic wavelength. Spectra represent the diagnostic electromagnetic absorption features of materials and thus it allows one to identify and quantify materials [2], [3]. A spectrum of each pixel in a hyperspectral image can be used for qualitative and quantitative analyses for minerals, and the results are then reflected based on each pixel to map the mineral types or the distribution of the specific mineral content in the scene. This noninvasive hyperspectral image analysis is much faster than the conventional chemical analysis and has different scales of varying applications in the geological remote sensing from space to the laboratory: remote sensing for mineral exploration and mine monitoring [4], [5], ground-based hyperspectral imaging to identify and quantify ore in a rock face [6], mineral mapping for drilling core [7], and ore grade classification and estimation [8], [9]. Hyperspectral imaging has huge potential to facilitate geoscience due to its mineral detection and content estimation capabilities. However, hyperspectral image analysis still has difficulties in detailed mixture quantification due to the spectral complexity, especially for ores composed of several minerals.

A considerable number of papers have been published on hyperspectral imaging for iron ore identification and quantification [10], [11], [12], [13], [14]. Hematite, the dominant mineral in iron ore, has a distinctive spectral absorption feature in the ultraviolet and VNIR wavelength ranges due to charge transfer [15], [16]. In addition, previous work has focused on diagnostic spectral absorption features such as absorption band, depth, and symmetry and developed prediction models that express the relationship between iron ore grade and those features. da Costa et al. [10] developed a linear regression model to predict iron ore grade from principal components of sample spectra. Thangavelu et al. [11] demonstrated a strong correlation between spectral absorption features of iron ore and iron ore grade, indicating feasibility in the prediction of iron ore grade via hyperspectral imaging. Murphy and Monteiro [12] mapped iron ore grade on a BIF vertical face by applying a linear regression model based on integrated derivative spectra as a spectral feature, while Noda and Yamaguchi [13] used iron ore absorption depth to quantify ore grade, and Rodger et al. [14] developed a specific index to quantify iron ore grade from absorption depth and width. Overall,

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these studies suggest a strong relationship between iron ore grade and its spectral features, demonstrating the capability of hyperspectral imaging in the prediction of iron ore grade by modeling the relationship between spectral features and ore grade. However, spectral features can vary depending on grain size, mineral component, mineral crystallinity, element substitution, and imaging conditions amongst others. Thus, extracting spectral features that represent the whole spectra in the image scene appropriately is a challenge. In addition, mineral ore is the intimate mixture of minerals, and thus, the relationship between spectral features and ore grade is nonlinear, which adds to the difficulty in attaining a correct prediction, linear regression model is usually applied, owing to the presumption of a linear relationship [17].

On the other side of traditional modeling, machine-learning-based methods have been developed to predict iron ore grades from spectral features more accurately [9], [18], [19], [20]. Machine learning is a computer algorithm that can perform a specific task by iteratively learning from large amounts of data [21]. Using a shallow neural network machine learning model, Xiao et al. [18] were able to estimate iron ore grade from visible–near-infrared (VNIR) spectra. In line with this, Rodger and Ramanaidou [19] used spectral features such as iron absorption depth and width to train a random forest regressor. Silva et al. [20] showed excellent prediction results with the machine learning based model for iron ore grade using spectral data. Nie et al. [9] demonstrated feature extraction by variational mode decomposition and made a random forest regressor using extracted spectral features. In essence, machine learning allows for the correct prediction of ore grade from spectra by modeling the intrinsic relationship between extracted spectral features and ore grade without presuming linearity or nonlinearity in the data. Although machine learning algorithms perform well in predicting ore grade, the proper spectral feature for the correct prediction differs amongst samples and the process of feature extraction depends on the analyst. Therefore, advanced automatic feature extraction methods that take into account all the spectral features are needed to achieve precise prediction.

As such, deep learning, an advanced machine learning method, has the potential to automatically extract and analyze input data features for a more accurate prediction due to its multilayered model architecture [22]. Convolutional neural networks (CNNs), a deep-learning-based model, have been used in numerous applications involving hyperspectral image analysis to classify and estimate ore grade because of their ability to automatically extract features [23], [24], [25], [26], [27]. CNN consists of multiple convolution layers, enabling it to extract data features without human intervention. Xiao et al. [23] developed a CNN-based model for five different grades of iron ore classification, while Okada et al. [24] presented the integration of hyperspectral imaging and CNN for mineral type and grain size classification. In addition, Prado et al. [25] used a 1-D and 2-D CNN to predict copper grade via spectra, while Xiao et al. [26] evaluated a combined model of 3D-CNN and Residual-Net for iron ore grade classification. Again, it should be highly emphasized that CNN can automatically extract input spectral features and model the nonlinear complex relationship between the extracted features and ore grade. Although CNN-based ore grade classification capabilities have been demonstrated by numerous researchers, little interest has been placed on ore grade prediction using hyperspectral image and CNNs. At the time of this research, none of them has focused on predicting iron ore grade via hyperspectral imaging.

Therefore, this article presents iron ore grade prediction using VNIR hyperspectral imaging and a CNN. Some machine learning and deep-learning-based iron ore grade prediction algorithms will be trained with various grades of iron ore samples for comparison, and model prediction performances will be evaluated. The trained models will then be analyzed to understand how the model determine ore grade.

II. Material and Methods

Iron ore grade prediction models based on machine learning and deep learning were developed by training the models with artificial iron ore mixture spectra, followed by evaluation of the models with natural iron ore spectra (Fig. 1). The trained deep learning model was then analyzed to visualize its interpretability.

A. Iron Ore Sample Preparation

Various grades of iron ore samples were prepared by mixing pure hematite, magnetite, and silica fine powder reagents to produce a balanced dataset for adequate training of the prediction model (Fig. 2). These reagents were mixed at a ratio of 10 mass% of each reagent in the range of 0–100 mass% and, a total of 66 samples were prepared. All the reagents were of high purity, specifically at 95%, 95%, and 99.9% for hematite, magnetite, and silica, respectively. All the mixture samples were homogeneously mixed using a porcelain mortar for 5 min. For validation purposes, natural iron ore samples were collected to test the generalizability of the prediction model. A total of nine iron ore samples were collected from several mine sites of eastern part of Japan, including the Wakasennin and Kamaishi mines in Iwate Prefecture and the Akatani mine in Niigata Prefecture; All the samples were formed in iron skarn deposits [28], [29], [30]. XRD (Rigaku MiniFlex) analysis revealed that the presence of hematite and magnetite in all the natural samples which is constant with the previous reports [28], [29], [30]. The collected samples were crushed with a jaw crusher and a vibrational mill to a particle size of less than 212 μm.

XRF analysis was conducted via a PANalytical Epsilon4 apparatus to quantify the chemical composition of all the samples including the mixture samples and the natural samples and the measured iron content of each sample was used as a target value in the model training. As shown in Table I, the natural samples are mostly composed of iron and silica, with traces of aluminum and calcium. The aluminum may be present as iron substation in the hematite, whereas the calcium may originate from the limestone, as the ores formed in the skarn.

B. Hyperspectral Imaging

Hyperspectral images are high-information-density images containing continuous reflectance spectra in each pixel,

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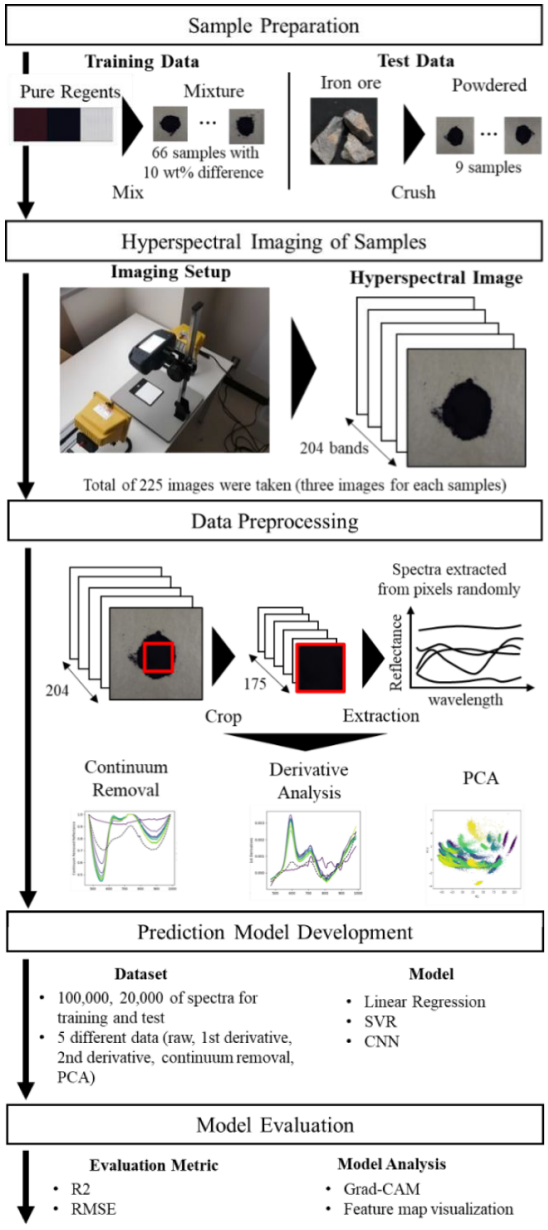


Fig. 1. Experimental flowchart.

TABLE I

CHEMICAL COMPOSITION OF THE NATURAL SAMPLES ANALYZED BY XRF (W, K, AND A REPRESENT WAKASENNIN, KAMAISHI, AND AKATANI, RESPECTIVELY)

	Fe2O3 (%)	SiO2 (%)	Al2O3 (%)	CaO (%)
W 1	77.563	14.629	0.953	0.099
W 2	62.73	36.206	n.d.	6.343
K 1	84.304	7.715	0.792	1.251
K 2	90.946	3.169	1.442	7.035
A 1	90.434	0.196	n.d.	0.485
A 2	82.191	8.613	4.78	0.53
A 3	98.072	0.368	n.d.	0.088
A 4	96.597	1.229	1.067	7.988
A 5	74.336	15.674	n.d.	6.271

information that can be used to identify and quantify material in an image scene nondestructively and noninvasively. In this work, hyperspectral images were captured from each of the

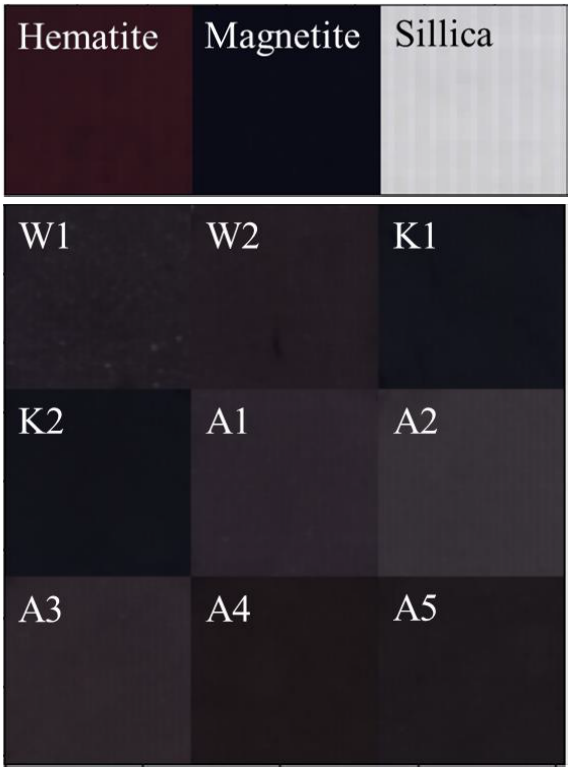


Fig. 2. False RGB images of powdered samples reconstructed from obtained hyperspectral images. In lower part, W, K, and A represent Wakasennin, Kamaishi, and Akatani, respectively.

75 samples with three images per samples via Specim IQ hyperspectral camera, and this is a push broom-type hyperspectral camera covering the VNIR range wavelength of 400–1000 nm, within which iron oxide absorption features are exhibited. It moreover covers this VNIR with a spectral resolution of 204 bands and a 512 × 512-pixel spatial resolution. The powdered samples were flattened to avoid illusion variation, followed by setting the samples about 30 cm away from the camera to ensure each sample was captured entirely; in addition, halogen illuminations that cover the VNIR region were used. The imaging setup is shown in Fig. 1. The obtained hyperspectral images were processed with a white and a dark reference to correct illumination and dark current effects, respectively. The exposure times were adjusted between the range of 3–25 ms relative to the sample's color, with lighter samples having a shorter exposure time, and darker samples with longer exposure time. The effect of the change in exposure time was corrected by dividing the reflectance by the exposure time.

C. Data Preprocessing and Feature Extraction

Hyperspectral data contain spectra reflecting the electromagnetic absorption characteristics of materials and allow for the classification and quantification of the samples based on spectral characteristics. Spectral features such as absorption band, width, depth, and symmetry of spectra are unique to materials and thus can be used for quantitative analysis. However, spectral features vary due to the sample conditions

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and measurement setups resulting in difficulty in the analysis of spectra, as such spectral analysis involves preprocessing of hyperspectral data [17], [31].

To correct spectra, continuum removal and derivative analysis were applied to the obtained spectral data. Continuum removal is a common spectral preprocessing technique used to correct baseline slopes in geological materials [3], [31]. This method assumes a background absorption called a continuum and derives a continuum as a convex hull of a spectrum. The continuum is removed from an original spectrum by dividing an original spectrum by the continuum. Derivative analysis differentiates original spectra and transforms original spectra into derivatives. The first derivative spectra correct baseline shift, and the second derivative spectra correct baseline shift and slope [3], [32]. Derivative analysis is sensitive to spectral noise, and thus usually performs with Savitky–Golay smoothing to alleviate a decrease in the signal-to-noise ratio which strongly affects the differentiation results [32]. In this work, Savitky–Golay smoothing was applied with the window size of 11, polynomial order of 2, and derivative of 1 and 2.

Principle components analysis (PCA) is performed to extract spectral features. PCA is used to reduce the dimensionality of hyperspectral images from the perspective of mathematical statistics [1], [3], [33]. This method transforms hundreds of bands of spectral data into a few informative feature vectors called principal components while keeping the high-density information.

D. Prediction Model Development

Modeling a relationship between spectral features that occur within specific chemical compounds and the chemical compositions of samples has been applied to predict ore grade in hyperspectral image analysis. Modeling often begins by making a dataset to train models which include data preprocessing and feature extraction. Thereafter, prediction models are trained using the dataset comprising the extracted spectral features and sample ore grade information to predict iron ore grade. In this work, we compare the conventional models including linear regression and support vector regression (SVR), which involve manual feature extraction and a deep-learning-based model, CNN, which extracts features automatically and evaluate the performance of each model. All model performance was evaluated with R^2 and root mean-squared error (RMSE) which indicate how well the model fits the data and the prediction error, respectively. The whole process was done in Python 3.8.

1) *Dataset*: The dataset containing different grades of iron ore spectra was created to train and test the prediction models. Spectral data were obtained from the acquired hyperspectral image by cropping the sample area from each image scene and extracting spectra from the pixels. A total of 100 000 spectra were randomly extracted from the hyperspectral images of each artificial mixture sample in fixed numbers for the model training. The extracted spectra were randomly split into two, 80% and 20% for training and validating the model respectively. The training dataset was used to train the model, while the validation dataset was not used for the model training to obtain unbiased evaluation. A total of 20 000 spectra were randomly extracted from the hyperspectral images of each natural iron ore sample in fixed numbers to test the trained model performance on the actual samples. The validation and test datasets were not used for model training. In addition, spectral bands ranging from 25 to 200 (466–991 nm) of the 204 bands were used for further analysis as camera sensor sensitivity outside this range causes noise. After the spectral extraction, the dataset undergoes preprocessing to remove the baseline shift and slope by continuum removal and derivative analysis before machine learning. As for the deep learning model, the dataset did not undergo this preprocessing stage due to its advanced processing capabilities.

In this work, four different spectral features, namely, integrated continuum removed spectra; integrated first derivative spectra; integrated second derivative spectra; and principal components were used to train machine learning models. Integration in this regard relates to a summation of the entire range of spectral bands representing an area of absorption exhibited by iron oxide. The first four principal components were selected as spectral features based on the number of pure samples and the contribution rate as will be discussed in the results section.

2) *Machine Learning Model*: Machine learning is a computer program that is able to improve the performance index of a specific task from input data in an iterative way. Machine learning has been widely used in geological classification and prediction tasks via hyperspectral imaging. In this work, linear regression and SVR were used to predict iron ore grade from spectral data. Linear regression is fundamental machine learning model that assumes a linear relationship between input features and object variables. This linear modeling method fits a linear model with weights of features by minimizing the residual sum of squares between object variables and the predicted value by the linear approximation from input data. The model was optimized by minimizing the mean-squared error between true iron ore grade and predicted iron ore grade from spectral features with the trained model.

SVR is an extended method of machine learning used to solve regression problems [34]. SVR has great applicability due to its ability to allow the model to capture the nonlinearity of the data [35]. This method aims to find a regression line (function) that approximates the relationship between input features and object variables with a kernel function. Fitting the regression model is achieved by mapping the input features to a high-dimensional feature space and finding the hyperplane that minimizes the prediction error. The training parameters of the SVR model were determined experimentally. The kernel function was set as radial basis function. The rest of the parameters were set to default based on Scikit Learn's library as the performance produced high accuracy results.

All the machine learning models were trained using the training dataset that consists of the preprocessed spectral features as the input and the corresponding iron ore grades' information as the object variable.

3) *Deep Learning Model*: Deep learning is one of the machine learning methods characterized by its multilayered model architecture and remarkable performance [36], [37]. CNN, the model used in this article, is a deep learning

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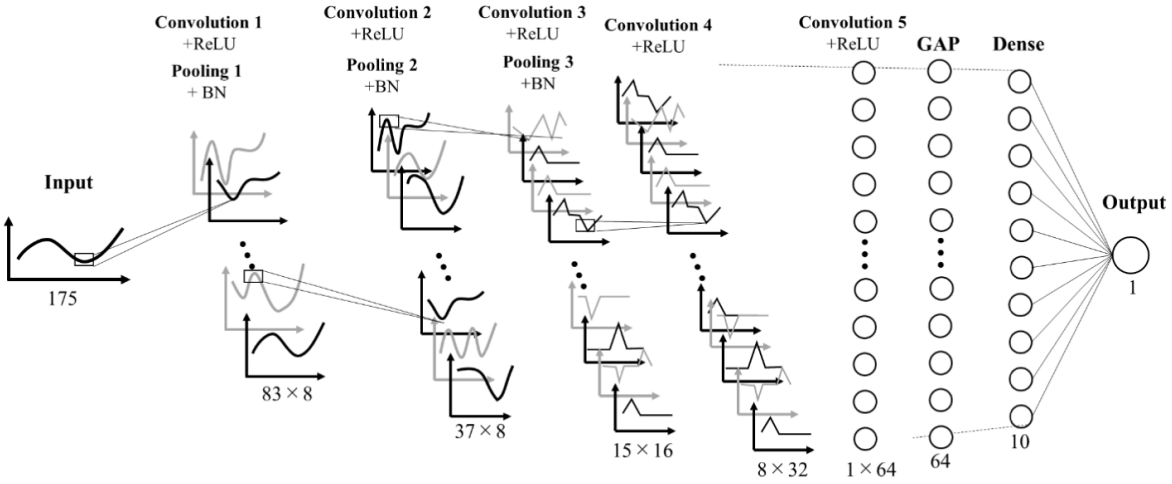


Fig. 3. Architecture of the 1-D CNN. The numbers below each layer and BN represents the output size at that layer and batch normalization respectively.

model consisting of a multitude of convolutional and pooling layers which automatically extract and compress input data features and achieve task accuracy by modeling nonlinear relationships between extracted features and object variables [36], [37]. In CNN model training, the input data are fed into the convolution layer where filtering process to extract important features for prediction is performed by moving a kernel over the spectrum with a specified stride and then outputting feature maps. The convolution feature maps pass through a pooling layer via a rectified linear unit (ReLU) activation function, which outputs the input value as positive, and outputs zero elsewhere, this is not the case, hence capturing the nonlinearity of the data. Pooling is applied as a way of data compression for faster model training while retaining the most important information from the data. Maximum pooling works by extracting the maximum value in a pooling window defined by the pooling size and stride from the input data. After convolution and pooling processes, the extracted features are then summarized at the dense layer where the CNN adequately models the relationship between extracted features and object variables for either classification or prediction tasks precisely at the output layer.

In this article, the 1D-CNN, which can handle the 1-D-spectral data, was built to predict iron ore grades from spectral information contained in the hyperspectral image. The decision to use 1D-CNN was based on its modeling and automatic feature extraction capabilities for spectral data, in contrast to the conventional machine learning which requires manual feature extraction. The 1D-CNN comprises an input layer, five convolution layers with the ReLU activation function, three max pooling layers with batch normalization to improve the generalization of the model, a global average pooling (GAP) layer for faster training, followed by a fully connected dense layer and an output layer as shown in Fig. 3. The Adam optimizer was used to compile the CNN with mean squared error between the predicted and true iron ore grade as loss. The CNN model was trained with the raw spectra dataset since convolution is assumed to work as a spectral preprocessor. In addition, the model was carefully trained

TABLE II
HYPERPARAMETERS OF THE 1-D CNN

	Filters	Kernel size	Stride	Padding
Conv 1	8	10	1	same
Conv 2	8	8	1	same
Conv 3	16	8	1	same
Conv 4	32	8	1	same
Conv 5	64	8	1	same
	Pool Size		**Stride**	
Max Pooling 1	5		3	
Max Pooling 2	5		3	
Max Pooling 3	5		3	
	Type	**Learning Rate**	**Epoch**	
Optimizer	Adam	0.001	1,000	

to avoid overfitting by monitoring the learning curve. The training dataset was separated into training and validation in a ratio of 8:2. The model's hyperparameters were determined experimentally as shown in Table II. All CNN modeling implementations were done using TensorFlow 2.1.0 in Python.

4) *Model Evaluation*: The model performance was evaluated with R^2 and RMSE. The R^2 and RMSE scores are common metrics to assess a model's ability to fit data and a prediction error, respectively. These two metrics was calculated on the both validation and test datasets to evaluate model generalizability and practicability, respectively. As the validation dataset was not used to train the model, the evaluation results of the validation dataset provide an unbiased assessment of whether the model is overfitted to the training dataset. The test dataset was composed of natural iron ore, and thus the evaluation results of the test dataset provide whether the model can be applicable on the real application cases.

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Grad-CAM and feature map visualization were used to analyze the trained deep learning model [38]. Grad-CAM can highlight the important part of the data for prediction, and thus it is a valuable analysis technique for understanding a deep learning model, which is often referred to as a black box, meaning it lacks explainability [38]. Grad-CAM computes a positive feature-importance map based on the feature map gradient of the last convolution layer. In this experiment, Grad-CAM was applied for the fourth layer rather than the fifth layer because the output size of the fifth layer was 1 which cannot be used for interpretation. In addition, the Grad-CAM output was linearly interpolated to specify the important bands out of the 175 bands for iron ore grade prediction; as such, the Grad-CAM output size was 8, which depended on the 4th convolution layer output size. Although Grad-CAM aides in explaining the basis of the model prediction by visualizing the feature importance map, how the model works remains unclear due to its deep layered architecture. To understand the feature extraction process of the trained deep model precisely, feature maps for each convolution layer were obtained. Grad-CAM and feature map visualization were conducted on the three pure endmember sample, hematite, magnetite, and silica spectra to verify that the model correctly extract the input spectra features.

III. Results and Discussion

A. Spectral Characterization of the Samples

Hyperspectral imaging of the various grades of iron mixture samples and the natural iron ore samples was conducted to develop ore grade prediction models. The average spectra of the pure and natural samples are shown in Fig. 4(a) to characterize the spectral features which should be the basis of the model prediction. All the sample spectra showed a moderate upward trend toward longer wavelengths. For the pure sample average spectra, it is apparent that they have different spectral signatures for each. The hematite average spectrum exhibited moderate reflectance, whereas the magnetite and the silica average spectra exhibited no distinct absorption and relatively lower and higher reflectance than that is hematite, respectively. The low reflectance of the magnetite spectrum caused high absorption coefficients over the VNIR range [39]. In addition, the hematite exhibited a week absorption caused by iron oxide at around 550 and 850 nm, while the magnetite and silica exhibited no distinctive absorption feature in the VNIR range [15], [16]. In this experiment, the magnetite spectral absorption occurring at around 550 nm and 1000 nm could not be found in the average spectra, which might be due to the sensitivity of the imaging sensor; even with the ten times longer exposure time it did not capture that absorption. For the natural sample average spectra, the Wakasennin and Akatani samples showed spectral signatures similar to that of the hematite spectrum which are moderate reflectance level and weak absorption at around 550 and 850 nm. In contrast, the Kamaishi samples showed spectral signatures similar to that of the magnetite spectrum which are lower reflectance level and no absorption. This suggests that the Wakasennin and Akatani samples and the Kamaishi samples are mainly composed of hematite and magnetite respectively. In addition, the Wakasennnin and Akatani samples presented the flatter spectra than hematite, and it could be explained by the presence of hematite with magnetite and silica. Likewise, the higher reflectance of the Kamaishi sample than magnetite could be due to the presence of magnetite with silica.

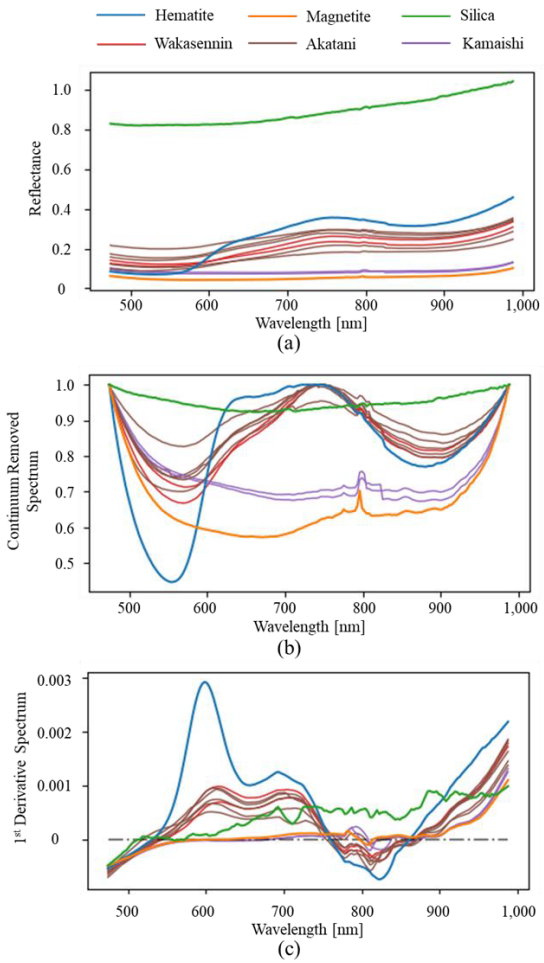


Fig. 4. Spectra of pure samples and natural samples (a) average reflectance spectra, (b) continuum removed average spectra, and (c) first derivative average spectra.

Continuum removal was applied for the average spectra to isolate the spectral feature caused from the background absorption. As shown in Fig. 4(b), all the continuum removed spectra had sharp peak at 800 nm. This peak can be interpreted as the measurement noise as it was present in all the spectra; the spectra with lower reflectance level had the stronger peaks due to the division of the continuum. For the pure samples, the continuum removed spectra clearly highlighted the distinct spectral absorption of the hematite. On the other hand, the continuum removed spectra of the magnetite and the silica resulted in U-shaped curves due to the flat continuum deduced from the featureless original spectra. For the natural samples, the continuum removed spectra of the Wakasennin and the Akatani samples showed spectral absorption centered at around 550 and 850 nm. While the absorption depth at 850 nm is almost the same as that of the hematite, the depth

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of absorption at 550 nm is shallower in the natural samples. This could be attributed to aluminum substitution, which makes the spectra more reflective within a certain wavelength range [40].

Derivative analysis was applied to remove baseline shift effect [Fig. 4(c)]. For the pure samples, the absorption spectral trough minima were found at near 550 and 850 nm for the hematite sample and at near 550 nm for the magnetite sample, which is in line with the previous research [15], [16]. For the natural samples, the Wakasennin and Akatani samples and the Kamaishi samples showed a similar trend to the hematite sample and the magnetite samples, respectively. Furthermore, subtle local minima shift toward longer wavelength was observed in the both the Wakasennin and Akatani samples at around 850 nm. This shift could be explained by aluminum substitution in the hematite [40].

In general, all the pure samples exhibited different spectral features; the hematite sample had moderate reflectance and apparent absorption at 550 and 850 nm; the magnetite sample had low reflectance and slight absorption at around 550 nm; and the silica sample had high reflectance. In addition, the natural samples presented a similar spectral tendency to the pure samples with slight differences caused by aluminum substitution and the sample composition

The different grades of mixture sample average spectra are shown in Fig. 5 to visualize the relationship between the endmember composition and the spectral features. The hematite spectrum can remarkably differ depending on the content of magnetite and silica. Fig. 5(a) illustrated that the hematite spectra gradually become flatter and less reflective as the magnetite content increases. Although the spectra of hematite and magnetite mixture differ completely depending on their composition, the iron content varies only 2% at most which could make training the model difficult. In addition, the baseline of hematite shifted upward as the silica content increased [Fig. 5(b)]. Interestingly, the reflectance of the hematite–silica mixture samples spectra was drastically lower than that of the pure silica sample, which might affect the performance of the model prediction. Similar trends were found in the magnetite–silica mixture [Fig. 5(c)]. This is mostly due to the high probability of encountering a dark grain, causing the reflectance to decrease much more than its weight fraction. Given these spectral mixing relationships, the natural sample spectra could be interpreted as follows; the Wakasennin and the Akatani sample spectra, which are similar to the hematite spectrum, become flatter and shifted upward due to the presence of magnetite and silica; the Kamaishi sample spectrum, which is similar to magnetite spectrum, shifted upward due to the presence of silica. Overall, the prediction model should account for these complex spectral changes depending on the sample composition and thus the nonlinear modeling is needed for robust accurate iron ore grade estimation.

Principal component analysis was applied to the mixture spectral data to reduce the data dimensionality while retaining the high-density data information. As shown in Fig. 6(a), the cumulative explained variance showed that the first four principal components accounted for more than 99% of the

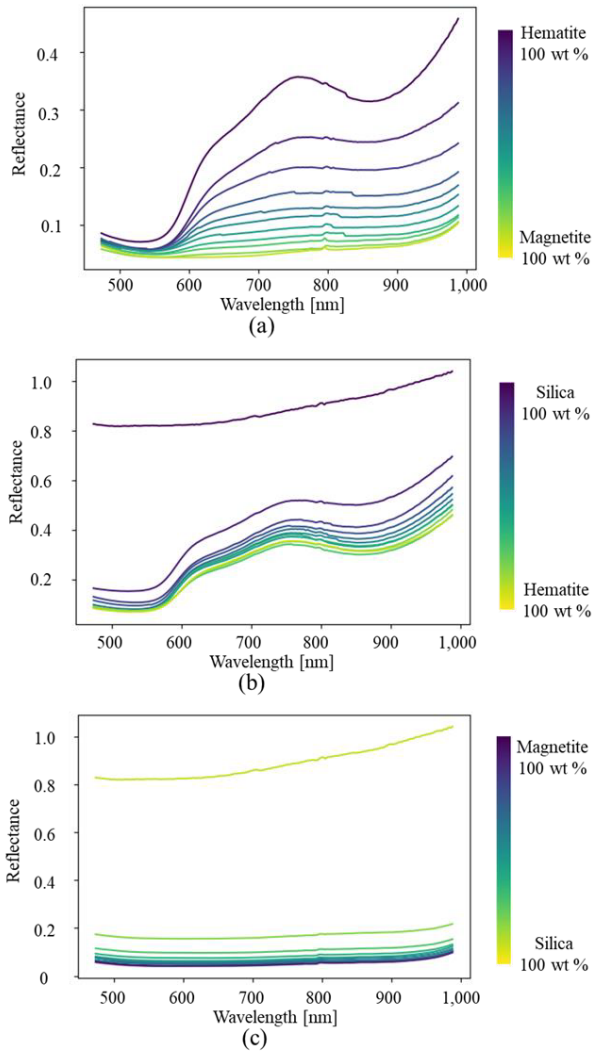


Fig. 5. Reflectance spectra of intimate mixture of pure samples illustrating the nonlinear aspect of reflectance spectra of mixtures. (a) Hematite and magnetite, (b) silica and hematite, and (c) magnetite and silica.

spectral variance, implying the majority of the spectral data can be explained by the first four principal components. The loading of the PC1 presented a flat shape with a moderate upward trend toward longer wavelength, whereas the loading of the PC2 presented troughs at around 550 and 850 nm. This could be interpreted as PC1 and PC2 representing magnetite and silica spectral features and hematite spectral features, respectively, from their resemblance. The principal components' plot showed that the data distribution differed between each mixture sample, with no significant correlation with the iron content. This irregular data distribution may necessitate the use of nonlinear modeling for accurate ore grade prediction.

B. Trained Model Prediction Results

Machine learning and deep learning models were trained with the training dataset containing the various grades of artificial iron ore spectra extracted from the hyperspectral images to predict the iron ore grade. The prediction performance of

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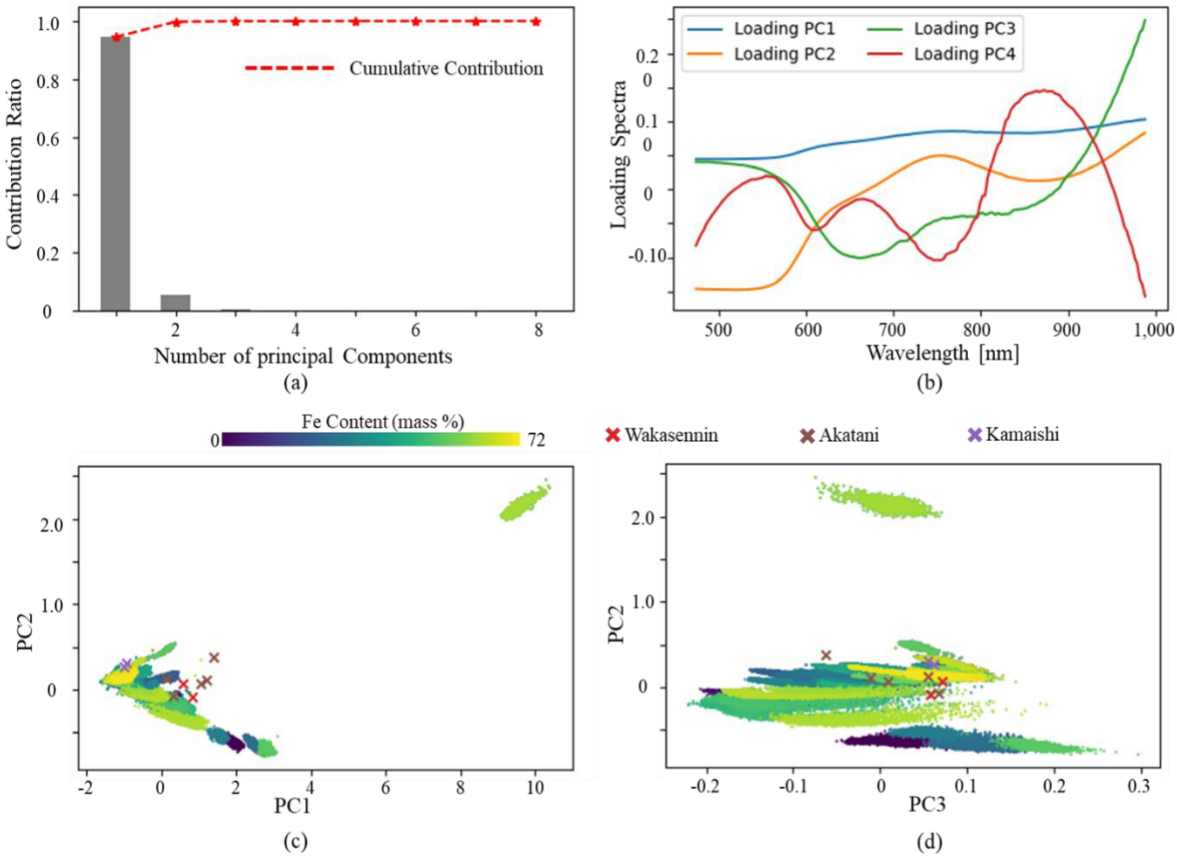


Fig. 6. Principle component analysis of the sample spectra (a) cumulative explained variance, (b) loadings of each principal component, (c) principal component 1 versus principal component 2, and (d) principal component 2 versus principal component 3.

TABLE III
EVALUATION RESULTS OF THE TRAINED MODELS

	Preprocessing			Validation		Test	
	Continuum Removal	PCA	Derivative	R^2	RMSE [mass%]	R^2	RMSE [mass%]
Linear	-	-	-	0.638	10.282	-3.556	15.964
Linear	✓	-	-	**0.686**	**9.574**	-6.579	20.590
Linear	-	-	1st	0.041	16.740	-1.229	11.166
Linear	-	-	2nd	0.077	16.423	**-1.024**	**10.641**
Linear	-	4	-	0.331	13.983	-1.866	12.661
SVR	-	-	-	**0.849**	**6.643**	-5.586	19.194
SVR	✓	-	-	0.741	8.629	-3.182	15.295
SVR	-	-	1st	0.179	15.491	-0.614	9.503
SVR	-	-	2nd	0.010	17.010	**-0.185**	**8.145**
SVR	-	4	0	0.825	7.152	-8.745	22.347
CNN	-	-	-	**0.973**	**2.757**	**-1.243**	**11.201**

these models was evaluated on the validation and test datasets using two metrics: the R^2 score to measure the model fit and the RMSE to measure the model accuracy. The metric scores of the validation dataset provide an unbiased evaluation, while those of the test dataset provide an assessment of the model practicability.

The results of the model training are shown in Table III. For the validation dataset, the linear model achieved the best scores of R^2 0.686 and RMSE 9.574 with the continuum removed dataset, while the SVR model achieved the best scores of R^2 0.849 and RMSE 6.643 with the unprocessed dataset. Overall, the SVR model surpasses the linear model, indicating the significance of nonlinear modeling for ore grade estimation. However, the results suggest that there is still room for improvement in feature extraction, as the model performances were significantly differed from the type of preprocessing. The low R^2 and RMSE scores for the derivative dataset could be explained by the baseline shift correction of the first derivative, which makes identifying magnetite and silica difficult. Thus, it could be concluded that proper feature extraction is challenging. This could suggest the difficulty in proper preprocessing and feature extraction for modeling. On the other hand, the CNN model achieved R^2 0.973 and RMSE 2.757, which are the best scores among all the experiments for the validation dataset. This excellent result can be because of the CNN model's high capability for automatic feature extraction and nonlinear modeling. These results can imply that the CNN model appropriately learns the relationships between mixed spectra and iron ore grade from the proper dataset.

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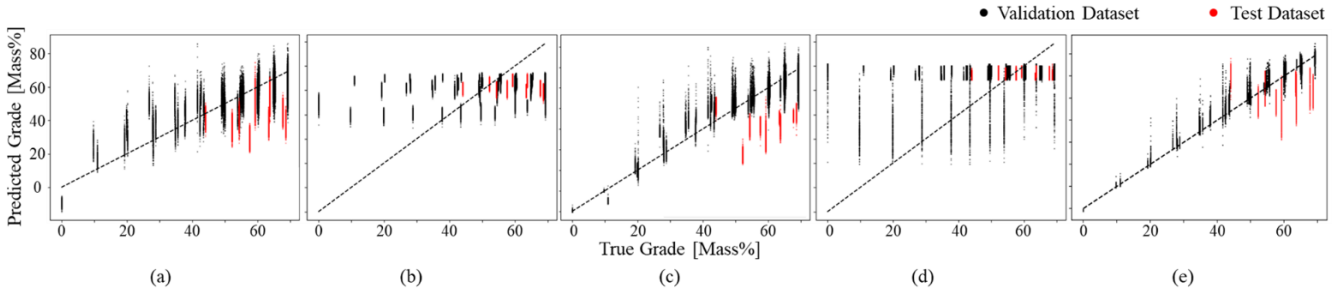


Fig. 7. Prediction results of the best score models (a) linear model trained with continuum removed spectra, (b) linear model trained with second derivatives, (c) SVR model trained with raw spectra, (d) SVR model trained with second derivatives, and (e) CNN model.

For the test dataset, the linear model showed the best scores of R^2 – 1.024 and RMSE 10.641 with the continuum removed dataset, while the SVR model achieved the best scores of R^2 – 0.185 and RMSE 8.145 with the second derivative dataset. The CNN model showed R^2 – 1.245 and RMSE 11.201 for the test dataset. In comparison of evaluation metrics scores for the test dataset among the models, the SVR model exhibited relatively better results with the first and second derivative datasets. Although the derivative dataset showed an exceptional fit with the test dataset, it cannot be used for reliable prediction since R^2 and RMSE scores for the validation dataset are fairly negative, meaning the models do not represent the relationship between the spectra and iron ore grade appropriately. In addition, the prediction results of the derivative dataset were concentrated at around 55–60 mass% as can be seen in Fig. 7. This biased prediction distribution was similar to the iron ore grade distribution of the test dataset, resulting in relatively better prediction performance with the derivative dataset. The R^2 scores were less than zero in all the cases, suggesting that the prediction performance of all the trained models for the test dataset was lower than random prediction. This low prediction performance could be attributed to the difference between the training and test datasets. The training dataset was composed of the artificial iron ore sample spectra showing how the spectra are changing along the endmember composition, while the test dataset was composed of the natural ore sample spectra which could represent compositional spectral changes and the spectral effect by the aluminum substitution in hematite and the grain size heterogeneity. Since the model was trained on the training dataset, it only learned the relationship between compositional spectral changes and iron ore grade, which could lead to lower prediction performance on the test dataset.

The CNN model, involving high capability for automatic feature extraction and nonlinear modeling, achieved the best R^2 and RMSE on the validation dataset among all the experiments. These magnificent achievements on R^2 and RMSE on the validation dataset can be interpreted that the trained CNN model properly learned the complex nonlinear relationship between the spectral features of the controlled grades of iron ore sample spectra and their grade for robust prediction than the conventional models. This indicates that the CNN model presented relatively higher generalizability than the conventional models. On the other hand, the metric scores

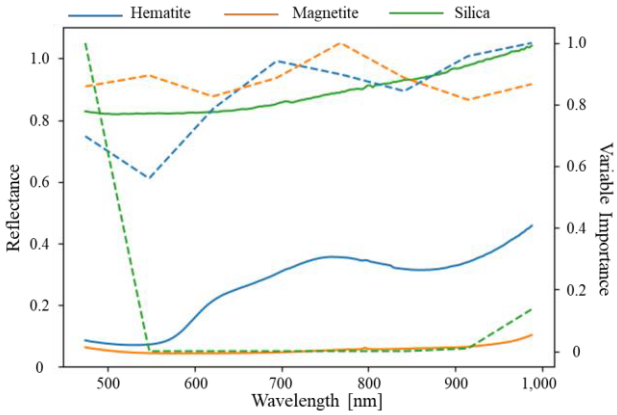


Fig. 8. Grad-CAM result visualizing feature importance map. The solid lines and the dashed lines show the average spectra of pure sample and Grad-CAM results, respectively.

for the test dataset were poor. This can be explained by the fact that the CNN model did not learn the spectral effect by the aluminum substitution and the grain size heterogeneity, as these were not included in the training dataset. Despite that low performance on the dataset, the model shows promise as it demonstrated the better learning capabilities than other models. Overall, the CNN model demonstrates its accurate prediction performance on the learned data, and thus it can be speculated that the CNN model will perform well on the natural sample once the model trained with more complex dataset.

C. Visualization of CNN Model

The trained CNN model was analyzed using Grad-CAM and feature map visualization to deduce the inner workings of the black-box CNN model, focusing on feature extraction process. Grad-CAM is a feature importance mapping method for the CNN model and highlights important features for prediction. Fig. 8 provides the variable importance for prediction obtained from Grad-CAM on the pure endmember samples spectra to evaluate whether the model pays appropriate attention to the spectral features. The variable importance for the pure samples showed the different trends for each sample. This could be inferred that the model identifies the endmembers from their individual features. The variable importance of

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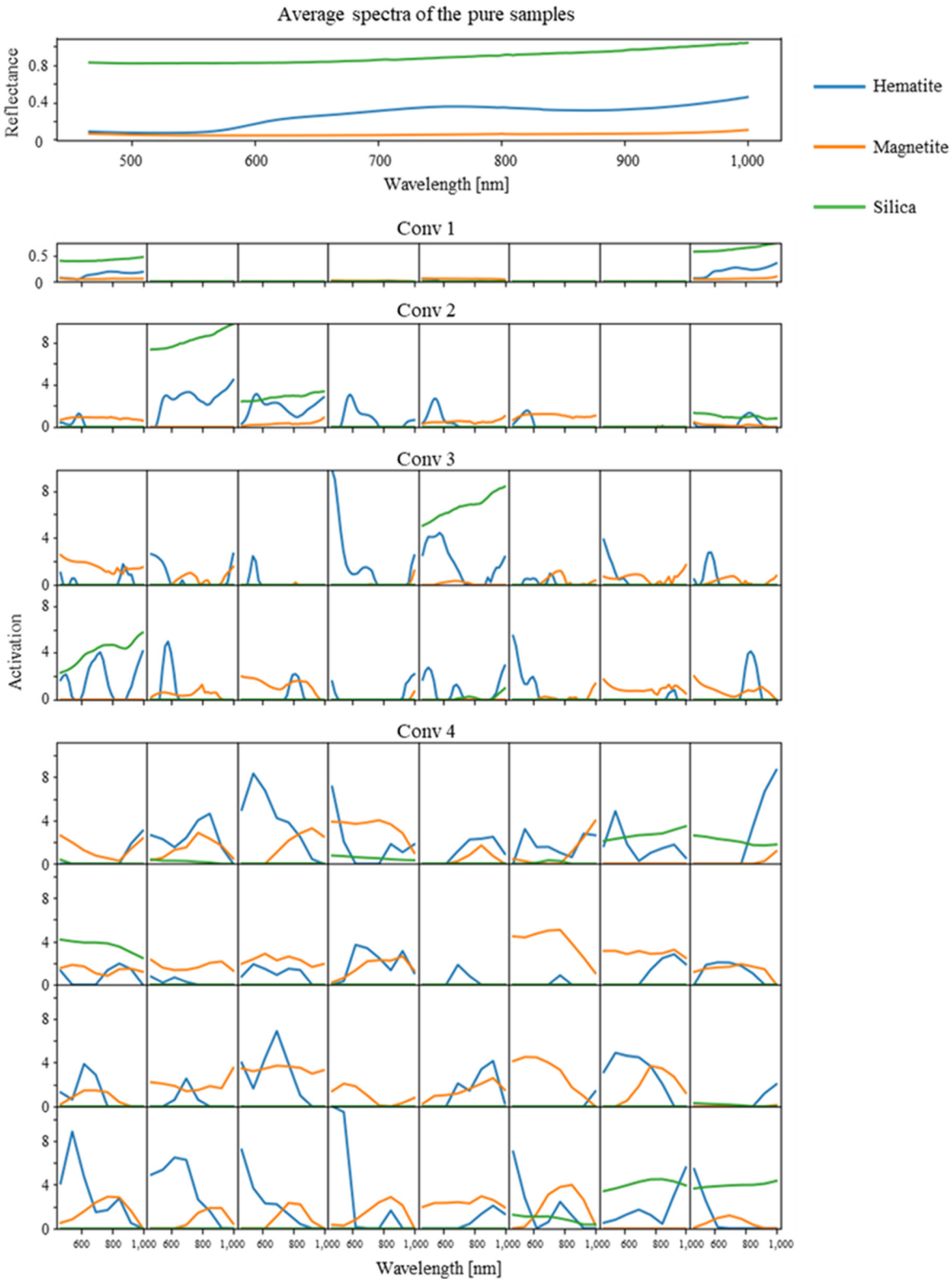


Fig. 9. Feature map visualization of each layer illustrating the feature extraction process. The top section shows an input spectrum (hematite, magnetite, and silica), and the following are the feature maps of Conv 1, 2, 3, and 4 from top.

hematite presented the troughs at around 550 and 850 nm which is constant with the spectral features of hematite. The variable importance of magnetite presented relatively flat trend with a high importance level over the entire wavelength range, which might be due to the featureless low reflectance spectra of magnetite. In addition, the highest importance variable was found at around 800 nm in the magnetite Grad-CAM results. Despite this minor peak could be interpreted as the effect of noise on the spectra, this peak may have no significant effect on the prediction, given that the importance level is almost the same for the other wavelengths and thus the CNN model seems to focus on the flatter shape of the magnetite spectrum.

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The variable importance of silica presented the higher values at both the edges, while it was insensitive to the others. This could be attributed to the gentle slope toward near-infrared of the silica spectra which likely to be explained only by the reflectance at both the edges. These results suggest that the CNN model can effectively extract the intrinsic spectral features that are unique to the samples to contribute to the high generalizability of the CNN model.

The feature map visualization was performed on the pure sample spectra to see the feature extraction process for each endmember (Fig. 9). The feature map is an output of a convolution layer, and the changes in the feature maps of each convolution layer illustrate the hierarchical intricate feature extraction mechanism of the CNN. The feature map by Conv 1 can be separated into two groups, nearly zero maps and the smoothed-like maps. The nearly zero maps could be interpreted as the insensitivity of the corresponding filters to the input spectrum features which can enhance the computation speed. The smoothed-like maps resemble the original spectra and can therefore account for the smoothing to reduce the noise in the input spectra. The following convolution layers can perform extraction and amplification of spectral features. The feature maps by Conv 2 exhibited some abstract peaks at around 550 and 850 nm in the hematite sample, and flat maps with different activation levels in the magnetite and silica samples. This could be explained by that the model extract diagnostic features of the samples which are the spectral absorption, low reflectance, and high reflectance for hematite, magnetite, and silica respectively. These results indicate that the second convolutional layers extract the informative spectral features with eliminating the nonvital information in the spectra. The feature maps by Conv 3 show the peaks and specific trends for each pure sample: the distinctive peaks located at around the beginning, the middle and the end of the diagnostic absorption bands on the hematite spectrum; the subtle peaks on the magnetite spectrum; and the nearly zero or upward trends on the silica spectrum. These results suggest that the reasonable features of the hematite and silica samples were extracted by Conv 3, while the subtle features of the magnetite, which appear sensitive to the noises, were extracted by Conv 3 due to the low reflectance level of magnetite. The feature map of Conv 4 yielded high activation at around characteristic absorption bands on the hematite spectrum, a peak at around 800 nm on the magnetite spectrum, and the flat and upward or downward trends on the silica spectrum. These feature maps seem to present almost the same tendency as the feature maps by Conv 2 and Conv 3. Hence, the fourth convolution layer resulted in an exaggeration of the extracted features by the second and third convolutional layers.

In summary, feature map visualization supports the idea that the trained CNN model can reasonably extract the global and local intrinsic spectral features associated with each pure sample throughout the convolutional processes, with no manual intervention. Therefore, it can be concluded that the trained CNN model can successfully predict the iron ore grade from spectral information with its automatic feature extraction and nonlinear modeling.

IV. Conclusion

Prior work has documented the effectiveness of deep learning models for classifying ore grades from a hyperspectral image. However, little attention has been paid to this deep-learning-based ore grade prediction for iron ore. This study thus evaluated the iron ore grade prediction performance of a deep learning model trained by comparing the conventional models and the CNN model.

The model evaluation results revealed that the CNN model achieved the best performance for the artificial iron ore samples indicating that the CNN model could automatically extract spectral features and learn the nonlinear relationship between spectral features and the corresponding iron ore grade. Grad-CAM and feature map visualization results supported this implication: Grad-CAM showed the important variable for the prediction that matched with the pure endmember spectra features, and feature map visualization illustrated that each convolutional layer worked to automatically preprocess, extract and emphasize the input spectral data appropriately.

Our study highlighted that the CNN can predict iron ore grade from hyperspectral images more accurately than the conventional models, which require the manual feature extraction, with nonlinear modeling for the prepared iron ore samples in the laboratory. Furthermore, we revealed that the CNN can adequately extract the spectral features which are unique to the sample without the individual intervention. The experimental results conclude that using a combined CNN and VNIR hyperspectral imaging system has an outstanding and robust prediction potential for the iron ore grade. This study is, therefore, of value to experts with an intent to implement hyperspectral imaging for accurate geological remote sensing from space to laboratory studies. However, although these results are positive, some limitations are worth noting, the major one being that our hyperspectral images were laboratory-sourced, meaning upcoming models would require retraining based on outdoor spectra in the case of outdoor ore grade delineation. In addition, the model was trained using the artificial mixture samples with a specific grain size distribution and no aluminum substitution. This means that the model could not learn the effect of the grain size and aluminum substitution on the spectra as can be seen with the prediction result of the test dataset. This suggests that the model should be trained using a more diverse dataset. Moreover, future work in the validation of applicability for ground-based and wide-area remote sensing presents research opportunities.

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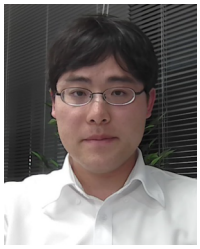
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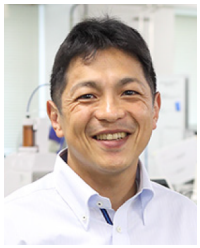
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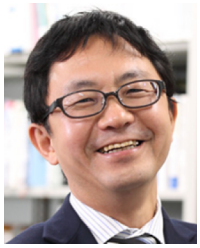
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